



Essentials of Prompt Engineering

Introduction

Why Prompt Engineering Is Important:

- Prompt engineering is essential because as generative AI and foundation models grow more powerful, the quality of the prompts directly affects the relevance, accuracy, and usefulness of the output. Well-crafted prompts improve results, boost efficiency, and enable clearer communication between humans and AI systems.
- Explore techniques like zero-shot, few-shot and chain-of-thought prompting.

Why Effective Prompting Matters in Generative AI

- Improving your prompts is the quickest way to control and optimize the output of a foundation model based on your specific goals or context.
- Effective prompt strategies enhance model performance and safety, introduce domain knowledge without fine-tuning, and lead to higher-quality outputs through better inputs.

Elements of a prompt:

- Instructions
- Context
- Input data
- Output indicator

Example prompt

Prompt

Given a list of customer orders and available inventory, determine which orders can be fulfilled and which items have to be restocked.

This task is essential for inventory management and order fulfillment processes in ecommerce or retail businesses.

Orders:

Order 1: Product A (5 units), Product B (3 units)

Order 2: Product C (2 units), Product B (2 units)

Inventory:

Product A: 8 units

Product B: 4 units

Product C: 1 unit

Fulfillment status:

Negative Prompting

- Negative prompting involves instructing a model on what *not* to generate, helping steer outputs away from undesired content or behaviors.
- It is useful for avoiding outputs like hate speech, explicit content, or biased language, leading to safer and more appropriate model responses.

Examples of negative prompting:

Image Generation

•**Prompt:** *"Generate a futuristic cityscape at night, but avoid including any weapons, violence, or dystopian elements."*

Text Generation

•**Prompt:** *"Write a product review that is informative and helpful, but do not include any negative or offensive language."*

•**Prompt:** *"Summarize this article without using political bias or sensationalist language."*

Modifying Prompts

Inference Parameters

- Foundation models allow you to configure inference parameters – such as randomness, diversity, and length – to shape model responses, though available options vary by model.

Randomness and diversity

- **Temperature;** controls the randomness or creativity of the model's output. A higher temperature makes the output more diverse and unpredictable and a lower temperature makes it more focused and predictable. Temperature is set between 0 and 1.
- **Top P;** controls the diversity of the text by limiting the number of words that the model can choose from based on their probabilities.
- **Top K;** limits the number of words to the top k most probable words, regardless of their percent probabilities.

Modifying Prompts

Length

Maximum length

The maximum length setting controls the number of tokens a model can generate during inference, helping prevent overly long or infinite outputs; the ideal value depends on the task – shorter for summarization or translation, and longer for creative or open-ended generation.

Stop sequences

Stop sequences are tokens that tell the model to stop generating output, overriding the maximum length setting, and are useful for tasks with unpredictable output lengths, such as ending responses in conversational AI systems.

Prompt Engineering Techniques

- **Zero-shot prompting**

The user relies on the model's general knowledge and capabilities to understand and carry out the task without any prior exposure, or *shots*, of similar tasks.

- **Few-shot prompting**

Supplement the prompt with sample inputs and their corresponding desired outputs, effectively giving the model a *few shots* or demonstrations to condition it for the requested task.

- **Chain-of-thought prompting (CoT)**

To initiate the chain-of-thought reasoning process in a machine learning model, you can use the phrase "Think step by step."

- **Tree of Thoughts (ToT)**

Extends CoT by exploring multiple reasoning paths simultaneously. Evaluates intermediate steps before reaching conclusion "Consider multiple approaches"

Prompt Misuses and Risks

Types of prompt misuses and risk:

- Poisoning, hijacking, and prompt injection

Poisoning

- Poisoning refers to the intentional introduction of malicious or biased data into the training dataset of a model. This can lead to the model producing biased, offensive, or harmful outputs, either intentionally or unintentionally.

Hijacking and prompt injection

- Hijacking and prompt injection refer to the technique of influencing the outputs of generative models by embedding specific instructions within the prompts themselves.
- A malicious actor could craft prompts for a text generation model that contain harmful, unethical or biased content. By doing so, the model might generate similar harmful, unethical or biased text outputs.

Prompt Misuses and Risks

Exposure Risk in Generative Models:

- Exposure refers to the risk of sensitive or confidential data being revealed by a foundation model during training or inference, potentially leading to privacy breaches.
- For example, training on private customer data may cause the model to unintentionally disclose personal information in its outputs, compromising user privacy and trust.

Jailbreaking

- Jailbreaking is the act of bypassing an AI model's built-in safety and ethical constraints to access restricted functions or generate harmful content.
- It involves using specially crafted prompts to exploit vulnerabilities in the model's filtering mechanisms and override intended limitations.

Knowledge Check

Q1.

A company notices that a customer-facing generative artificial intelligence (generative AI) chatbot is inadvertently outputting sensitive personally identifiable information from its training data.

Which risk is being illustrated in this scenario?

- A. Jailbreaking: The chatbot is being "jailbroken" by prompts that trick it into bypassing its ethical constraints and content filters, permitting it to generate explicit or offensive content.
- B. Prompt leaking: The chatbot's prompts and conversational context are being "leaked" to external sources, potentially exposing sensitive information or allowing unauthorized access.
- C. Exposure: The model's sensitive training data has been leaked or made publicly accessible, leading to data leaks and privacy violations.
- D. Hijacking: Malicious actors crafted prompts that have persuaded the model to generate harmful, unethical, or biased text outputs.

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Q2.

Which option correctly describes the concept of *negative prompts* in prompt engineering?

- A. Negative prompts are additional instructions provided to the language model to guide it towards generating undesired outputs.
- B. Negative prompts are examples of incorrect or undesirable outputs that the language model should avoid generating.
- C. Negative prompts are context or background information that the language model should ignore while generating responses.
- D. Negative prompts are techniques used to reduce the dimensionality of the language model's latent space.

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Q3.

A data scientist at AnyCompany wants to influence a model's responses by adjusting various inference parameters to achieve the desired level of diversity, coherence, and relevance in the generated outputs.

Which statement correctly describes the effect of increasing the temperature parameter during inference?

- A. Increasing the temperature parameter will make the model's responses more deterministic and less diverse, favoring the most probable outputs.
- B. Increasing the temperature parameter will make the model's responses more diverse and less deterministic, exploring a wider range of possible outputs.
- C. Increasing the temperature parameter will have no effect on the diversity of the model's responses, because it only impacts the length of the generated outputs.
- D. Increasing the temperature parameter will make the model's responses shorter and more concise, because it reduces the probability of generating longer sequences.

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Q4.

AnyCompany is developing a generative artificial intelligence (generative AI) system to assist customers in filing their tax returns. The process of filing taxes involves several complex steps, such as gathering relevant documents, calculating income and deductions, and completing various forms and schedules.

Which approach to prompt engineering would be most effective for this tax filing assistance system?

- A. Use a single, comprehensive prompt that covers the entire tax filing process.
- B. Provide the language model with a detailed prompt that explains the tax filing process from start to finish.
- C. Rely solely on the language model's pretraining on tax-related data, without any additional prompt engineering.
- D. Break down the tax filing process into smaller, more manageable tasks, and craft separate prompts for each task.

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Q5.

An organization is using a generative model to solve complex mathematical word problems. The prompt used must break down the problem, identify relevant information, and apply appropriate problem-solving strategies and calculations.

Which prompt engineering technique would be most effective for this mathematical problem-solving?

- A. Zero-shot prompting: Provide the language model with a single prompt containing the word problem and ask it to solve it.
- B. Few-shot prompting: Include a small number of examples of solved word problems in the prompt.
- C. Chain-of-thought prompting: Prompt the language model to break down the problem-solving process into a series of logical steps and show its step-by-step reasoning.
- D. Chain-of-thought prompting combined with few-shot prompting: Prompt the model to break down the problem-solving process into a series of logical steps, and include a few examples of solved word problems.

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